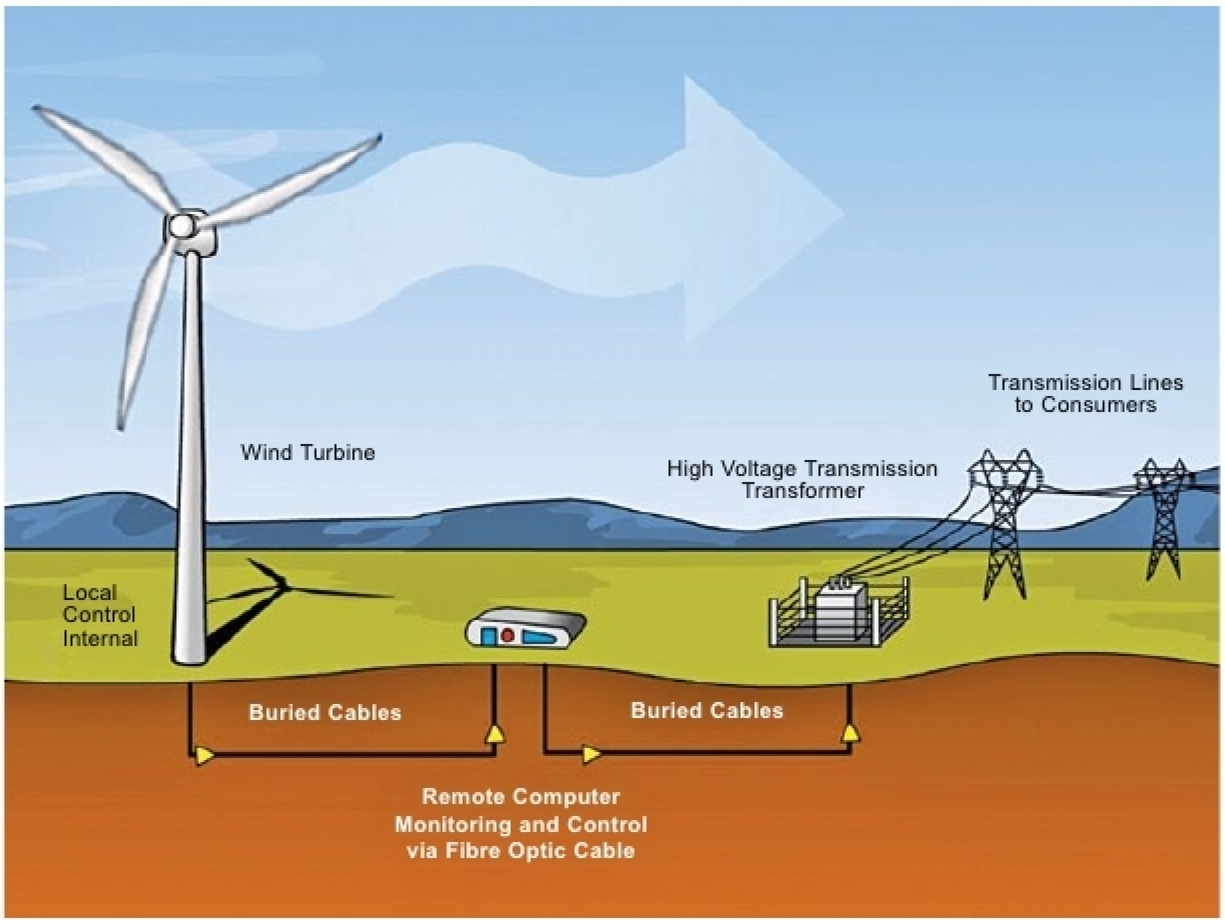
trash using smart waste collection systems that comprise sensors. Ultrasonic sensors monitor the capacity of dustbins from anywhere, providing detailed analytics to help track waste. This data can reduce fuel emissions by optimising trash pickup schedules. The technology also helps reduce the costs that come with overfilled dumpsters. Image-based trash-can sensors automatically monitor fullness and contents.

The technology optimises driver routes and fuel efficiency through GPS tracking, along with tilt monitoring, which records when a container gets picked up and put down. Smart waste technology helps with managing container inventory, build better routes and even accept orders through automated text messages. It determines which containers need to be serviced, schedules routes and evenly distributes jobs to drivers automatically.

A smart waste container automatically recognises, sorts and compresses waste using a camera, sensors and artificial intelligence (AI). It sorts plastic, paper and glass, based on the shape, colour and material within the container. Thereafter, waste is compressed so that total volume can be up to five times less. Once full, this self-sorting waste bin automatically notifies the trash collection company.

Wind energy monitoring. It includes high-performing monitoring solutions for wind data processing. It continuously monitors the vibration of wind turbines to detect faults at an early stage. Sensors are attached to gearboxes, generators and drive trains to provide real-time data streams that are transmitted to operators who monitor the results on a computer for changes that highlight a potential problem.

Data is easily retrieved via GPRS, the Internet and other methods. All complex mathematical algorithms are performed using cloud computing, rather than expensive sensors. This reduces the condition-monitoring unit to a simpler, more robust piece of equipment. The technology lets a wind farm transmit data collected from every sensor on its turbines, wirelessly or via the Ethernet, back to a single server for data processing. This helps achieve cost savings on sensor equipment and a more scientific approach to data analysis.



Wind monitoring system (Credit: Wikimedia Commons)

Today, wind farms offer owners a whole new way to approach operations and maintenance, thanks to condition monitoring and advanced prognostic systems. Condition monitoring helps them monitor the health of turbine components and related electrical systems. Its purpose is to predict maintenance issues so site operators can conduct repairs and replacements only when needed, to avoid unnecessary and costly uptower jobs.

Cloud-based software technology with cost-effective hardware can create reliable asset monitoring and predictive maintenance. The technology provides failure rate data of each individual turbine, along with a rolling five-year forecast into the predictive health of each asset at component and major system levels.

Some advanced environmental sensors are optical gas imaging cameras, sensors-on-a-chip technology, unmanned aerial vehicles (UAVs), advanced amplifiers, single-chip relative humidity, diffraction-based, micro sensors for ocean acidification monitoring, IR-based methane sensors for monitoring sub-sea methane releases, fluorescence technology to detect oil leaks at sea and nanostructure sensors for marine monitoring applications.

Design regulations

Health and safety services for hospitals, nursing homes, clinics, laboratories and general industry are developed and administered in strict compliance with current laws and regulations using equipment and techniques approved by Occupational Safety and Health Administration, National Institute of Occupational Safety and Health, National Fire Protection Association and American Industrial Hygiene Association.

Next-generation compliance consists of the following four interconnected components:

- Design regulations and permits to improve compliance and environmental outcomes
- Advanced emission and pollutant detection technology to measure pollutant discharges, environmental conditions and non-compliance
- Information tracking to help co-regulators better manage information, improve effectiveness and transparency
- Data analytics to achieve more widespread compliance EFY